Jonas Frerick

Curriculum Vitae

0000-0002-2778-3434

Keywords

Particle Physics, astrophysics, cosmology;
Light and Ultralight Dark Sectors, Dark Photons, Inelastic Dark Matter, Direct Detection, Self Interacting Dark Matter, Big Bang nucleosynthesis

Education

2021–2024 **PhD**, *DESY Hamburg*, Germany, GPA – 1.0 (magna cum laude)

Thesis *Catching Dark Photons in the Sky: Looking for light vector particles using satellites*

Supervisor Kai Schmidt-Hoberg

Description I studies different aspects of the Dark Photon model. The emphasis was on solar production of Dark Photons and direct gauged Dark Photon Dark Matter searches.

2019–2021 **M.Sc. Physics**, *RWTH Aachen*, Germany, *GPA – 1.0*
Focus of Studies: Quantum Field Theory and Gauge Theories

Thesis *Exothermic Dark Matter from Upscattering in the Earth*

Supervisor Felix Kahlhoefer

Description I built a general framework to calculate the effects of Dark Matter with two almost mass-degenerate states at direct detection experiments. (Led to publication [1].)

2016–2019 **B.Sc. Physics**, *RWTH Aachen*, Germany, *GPA – 1.3*

Thesis *Self-interacting Dark Matter from the Yukawa Potential: From astrophysical data to particle properties*

Supervisor Felix Kahlhoefer

Description I studied the effect of Dark Matter self-interactions on the shape of Dark Matter halos.

2008–2016 **Abitur**, *Qualification to attend university with focus on Physics and Mathematics*, *GPA – 1.0*

German system, *1.0 – 4.0*, 1.0 is the highest possible score (5.0 corresponds to not passed)

Publications

[1] Timon Emken, Jonas Frerick, Saniya Heeba, and Felix Kahlhoefer. Electron recoils from terrestrial upscattering of inelastic dark matter. *Phys. Rev. D*, 105(5):055023, 2022. `arXiv:2112.06930`, `doi:10.1103/PhysRevD.105.055023`.

[2] Jonas Frerick, Felix Kahlhoefer, and Kai Schmidt-Hoberg. A' view of the sunrise: boosting helioscopes with angular information. *JCAP*, 03:001, 2023. `arXiv:2211.00022`, `doi:10.1088/1475-7516/2023/03/001`.

[3] Jonas Frerick, Joerg Jaeckel, Felix Kahlhoefer, and Kai Schmidt-Hoberg. Riding the dark matter wave: Novel limits on general dark photons from LISA Pathfinder. *Phys. Lett. B*, 848:138328, 2024. `arXiv:2310.06017`, `doi:10.1016/j.physletb.2023.138328`.

[4] Sara Bianco, Paul Frederik Depta, Jonas Frerick, Thomas Hambye, Marco Hufnagel, and Kai Schmidt-Hoberg. Photo- and Hadrodisintegration constraints on massive relics decaying into neutrinos. 5 2025. `arXiv:2505.01492`.

Selected Presentations

Conference talks

iDM@IDM: Electron recoils from terrestrial upscattering of inelastic dark matter, IDM 2022, Vienna; 19.07.2022

A' view of the sunrise: Boosting Helioscopes with Angular Information , IAXO Collaboration Meeting, Hamburg; 15.03.2023

Photo- and hadron-dissociation constraints on relics decaying into neutrinos , Dark Matter and Neutrinos, Paris; 20.05.2023

Lectures

Quantum Computing V, DESY Workshop Seminar (winter term 22/23), Hamburg; 07.02.2023

Seminars

Looking for Dark Photon Dark Matter with LISA Pathfinder, Fermilab Theory Chalk Talk, Fermilab; 11.10.2023

Photo- and Hadrodisintegration constraints on massive relics decaying into neutrinos, BSM Seminar, KIT, TTP; 30.07.2025

Schools & Visits

1. *Summer School on Particle Physics*, ICTP Trieste, 19.06.2023-30.06.2023
2. *Asymmetry Secondment*, Fermilab, 20.08.2023-21.10.2023

Grants, Fellowships & Awards

2020-2022	Dean's List of the RWTH Aachen University	
2016	DPG Abiturpreis (Awarded by the German Physical Society DPG)	
English	Fluent	
German	Native	
French	Basic	*Basic words and phrases only*
Italian	Basic	*Basic words and phrases only*